

FACULTY OF ENGINEERING

Rule based expert System

For Career Guidance(Knowledge Represenatation)

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Short Introduction of your project (1-2 paragraph); what is your project, what is the main idea, why will be used, what is the goals

The project proposes a rule -based expert system that provides personalized guidance to users based on their interests , skills, and academic background. Instead of relying on large datasets and machine-learning models , the system uses if- then rules created with the help of domain knowledge from career counselling, job description and academic program requirements. By asking the user's a series of structured questions , the system can infer suitable career paths and recommend relevant fields to study or specialization.

Table Of Contents

1.Introduction	1
1.1 Problem definition	1
1.2 Goals	1
2. Literature Survey	2
3. Background Information	2
3.1 Required software	3
3.2 Other software	3
3.3 Hardware	3
4.1 First Module Name	4
4.1.1 Submodule name of first	4
4.2 Second Module Name	4
4.3 Third Module Name	4
5. Risk Analysis	4
6. Ethics.....	4
7. Conclusion	5
7.1 Benefits	5
a. Benefits to users :	6
b. Benefits to me :	6
7.2 Future Works	6
8. References	7

1.Introduction

1.1 Problem definition

Choosing a suitable career path is a complex decision that often requires expert guidance. Many student's and young professionals lack access to qualified career consultants or do not know how to systematically evaluate their interest, skills, an academic background. As a result , they may select programs or jobs that do not match their strengths leading to dissatisfaction , low performance , or frequent career changes systematic assessment of the ones own aptitude , interests ,and the realities of the job market

The Rule-Based Expert System for Career Guidance aims to address this problem by offering an automated, accessible and consistent form of the guidance. Using the knowledge representation and rule-based reasoning, the systems model typical relationships between interests (e.g analytical , artistic ,social) , skills (e.g programming, communication, problem-solving) and academic backgrounds (e.g sciences, engineering, humanities) and possible careers (e .g software engineer , nurse, teacher , graphic designer). When a user answers a questionnaire, the system evaluates the responses against its rule base and infers appropriate recommendations.

Unlike data – driven systems that require large datasets and intensive training , this expert system relies on transparent if-then rules . This makes the reasoning process easier to understand , modify , and validate . The project demonstrates how knowledge representation and inference can be used to solve a real life decision making problem in a structured way.

1.2 Goals

The main goals are:

- 1: To design and implement a rule-based expert system that provides career suggestions based on user input.
- 2: To apply knowledge representation techniques to encode career knowledge in the form of if – then rules.
- 3: To implement a single inference engine that evaluates user answers and matches them with relevant rules.
- 4 : To develop a user friendly interaction interface(command line or basic GUI) that guides the user through the questionnaire and clearly displays the results.
- 5: To demonstrate that such systems can be implemented in Python without large datasets,focusing instead on reasoning and inference.

2. Literature Survey

Several existing systems and websites offer career test or recommendation tools. Many of them are based on psychological models such as Hollands RIASEC types or use machine-learning algorithms that analyze historical user data and job trends. These systems usually require large datasets and may not clearly explain why a particular recommendation was made.

In contrast , traditional expert systems encode the knowledge of human experts in the form of rules. Classical expert system such as MYCIN in the medical domain have shown that rule-based reasoning can reach expert-level decisions when the knowledge base is well designed . For career guidance, a rule -based approach has the advantage of transparency each recommendation can be traced back to the rules that fired.

The proposed project differs fro typical online career quizzes by emphasizing explicit knowledge representation and inference . Rather than simply scoring answers with fixed weights , it uses logical rules such as :

. IF user is highly interested in programming AND has a strong analytical skills AND prefers working with technology

THEN suggest careers like “ Software Engineer” or “Nurse”.

Compared to data-driven recommender systems, this approach is easier to modify, does not require user data for training , and is suitable for educational purposes in an Artificial Intelligence or Knowledge Representation courses

3. Background Information

3.1 Required software

Python

The project will be implemented primarily in Python because it is easy to learn, widely used in AI and education, and has good support for rule-based logic through simple control structures or dedicated libraries

KIVY

Provides a user-friendly GUI, enhances interaction, supports cross-platform use, and fits well with Python.

SQLite

Offers lightweight, structured, persistent data storage that is easy to maintain, scalable, and required no external server.

3.3 Hardware

The system will run on a standard personal computer or laptop no special hardware is required beyond typical development hardware(CPU and RAM)

4. Modules

An online sales application may have a front-end for customers and a back-end for employees, just as an embedded system project may have both hardware and software components.

4.1 User Interaction Module

The end user interacts with the system through this module. It responds to prompts, shows queries, gathers responses, and displays the final career suggestions.

4.1.1 Questionnaire and Input Handling Submodule

This submodule defines the structure of the questionnaire: questions about interests, skills, and academic background. It validates user input, converts answers into internal representations (e.g., “high interest in programming”), and passes them to the inference engine.

4.2 Inference Engine Module

Rule-based reasoning is applied by the inference engine. After processing user input, it compares it to a list of if-then rules. The engine makes recommendations, including suggested career fields and brief explanations, when the requirements of one or more rules are met. Results may also be ranked by the engine according to the number of conditions met.

4.3 Knowledge Base Management Module

The career rules and associated data are managed and stored in this module. It includes the following:

- Career category definitions (e.g., technical, creative, social).
- Guidelines that relate backgrounds, interests, and abilities to professions.
- Optional descriptions or justifications for every career path.

5. Risk Analysis

Biased or incomplete knowledge base:

- The recommendations may not be fair or accurate if the rules favour particular fields or do not cover enough career options. Information should be gathered from a variety of sources, including job descriptions, academic advisors, and career websites, to be able to lower this risk
- Oversimplification of user profiles: Using straightforward rules to represent a complex human profile may overlook individual characteristics like values, personality, or local employment market conditions. The system will be positioned as a tool for support rather than a final decision maker, and this limitation will be recognised.

- Recommendation misinterpretation: Users may take the system's recommendations as a guarantee of professional success. The system will explicitly indicate that results are advisory and that users are urged to speak with human counsellors in order to mitigate this risk.
- Technical problems: Inaccurate or non-existent recommendations may result from errors in rule evaluation or input handling. To reduce this risk, appropriate testing with various user profiles will be carried out.
- Scalability: Managing and debugging the knowledge base may become difficult as more rules are added. As a result, the project will use a clear naming and grouping scheme for careers and rules.

6. Ethics

The project discusses individual choices that have a significant impact on a user's life. This results to ethical issues because:

- Recommendations from the system cannot be discriminatory on the basis of gender, race, religion, or other protected traits. Only academic background, interests, and skills will be considered in the rules. It is important to convey recommendations as suggestions rather than orders.
- Instead of mindlessly following the system, users should be urged to think critically and look for more guidance.
- User information, including responses about background or interests, should be treated with confidentiality and never kept on file without permission.
- Informed Consent and User Awareness: Users must be specifically informed of the system's capabilities and limitations before engaging with it.

7. Conclusion

The application of rule-based reasoning and knowledge representation to a practical decision problem is illustrated by the Rule-Based Expert System for Career Guidance. Without the need for intricate machine-learning models or sizable datasets, the system offers individualised career recommendations by posing a structured series of questions and using clear if-then rules.

7.1 Benefits

Personalised and Unbiased Results:

- Get suggestions that are free from human bias and are based only on your skills, interests, and values. Your profile is objectively matched to careers you might have missed by the system.

a. Benefits to users :

- Provides easy, quick, and accessible career guidance.
- Helps users understand the connections between their interests, skills, and possible careers.
- Clearly explains the rationale behind the recommendation of a specific career.

Benefits to us :

- Deepens our comprehension of rule-based systems and knowledge representation.
- improves our Python programming and software design skills.
- gives us practical experience in all aspects of project development, including requirements analysis, documentation, and implementation

Why did we chose this project?

We selected this project because choosing a career is a common problem for students, and we wanted to use an artificial intelligence concept to develop a tool that could provide useful assistance. The use of rule-based systems for career counselling combines technical expertise with a useful and practical application, making them a classic AI topic. It is instructive and personally inspiring to think of turning the knowledge of experts into rules and watching the system come to a conclusion.

7.2 Future Works

In the future, the system could be expanded with more detailed questionnaires, additional careers and localization for different countries and educational systems. To increase usability, a web application or graphical user interface could be included. In order to improve or add new rules over time, the rule-based system may also be coupled with data-driven techniques like gathering anonymous user feedback.

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